

Word Embeddings and Semantic Representations

Abstract

Word embeddings are dense vector representations of words learned from large text corpora. These representations capture semantic and syntactic relationships between words, enabling downstream NLP tasks to leverage rich linguistic structure with minimal supervision.

Background

Traditional bag-of-words representations treat each word as an independent symbol, ignoring the rich semantic relationships between words. Distributed representations address this limitation by mapping words into a continuous vector space.

Word2Vec

The Word2Vec model learns word vectors by training a shallow neural network to predict either the context words given a target word (Skip-gram) or the target word given its context (CBOW). The resulting vectors encode semantic similarity.

Applications

Pre-trained embeddings have been applied successfully to sentiment analysis, named entity recognition, question answering, and machine translation. Transfer learning from large corpora significantly reduces the labeled data required for fine-tuning.